

Stress-strain state of combined steel-FRP reinforced concrete beams

Ravshanbek Mavlonov^{1}, Sobirjon Razzakov¹, and Sohiba Numanova¹*

¹Namangan Engineering Construction Institute, 2J5J+26X Namangan, Uzbekistan

Abstract. Steel reinforcements in reinforced concrete structures are susceptible to corrosion under different exposure conditions. This can lead to some disadvantages, including concrete deterioration, reduced long-term service life, increased cost of the structure due to re-strengthening measures, and reduced overall durability of the structure. In order to solve these problems, the issue of comprehensive use of Fiber reinforced polymer (FRP) reinforcements as an alternative to steel bars is urgent. FRP reinforcements have specific advantages including corrosion resistance, high tensile strength, density four times lighter than steel, and also linear expansion coefficient under the influence of temperature is small like concrete. In order to increase the load bearing capacity and ductility, it is recommended to effectively use steel rebar together with FRP rebar as a combination reinforcement, taking into account brittleness characteristic of FRP reinforcement and low modulus of elasticity. In this article, concrete beams with combined reinforcement are modelled by using ANSYS Workbench 2022 software. By testing virtual model, deflection corresponding to the value of the applied load on the beam, compressive and tensile stresses in the concrete, and stresses in FRP and steel reinforcement located in the tension zone were determined and analyzed.

1 Introduction

Composite polymer materials are an alternative to traditional building materials and are gradually entering modern construction. Currently, volume of composite materials in the global market is close to 60 billion euros, and its worldwide production is more than 8 million tons [1,2,3].

A special place among composite materials is occupied by composite polymer reinforcement - this is a power rod made of continuous reinforcing roving and thermosetting resin. FRP bars are divided into glass (GFRP), basalt (BFRP), carbon (CFRP) and aramid (AFRP) FRP types [1-10].

Creation of composite reinforcement appeared due to the rapid development of chemical industry. In a number of countries, including the former Soviet Union, the United States, Germany, England, the Netherlands, Japan and other developed countries, a number of scientific studies on the creation of composite reinforcements began to be carried out in the 60s of the 20th century [1-3].

* Corresponding author: ravshanbek.mavlonov@gmail.com

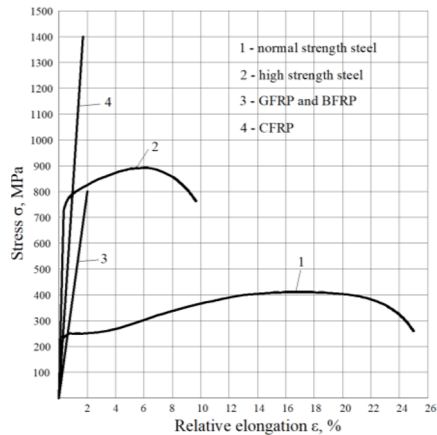


Fig. 1. Stress-strain curve of steel and FRP rebar

The stress-strain curve of CFRP, GFRP and BFRP under static loading is linear, no yield point and it breaks suddenly like brittle materials, and the relative elongation is 1.7 for CFRP and 2% for GFRP and BFRP. Relative elongation is up to 25% for normal strength steels and 9% for high strength steels. However, if the strength of CFRP is up to 1400 MPa, it can be seen that it is 800 MPa in GFRP and BFRP, this indicator is 400 MPa in normal strength steels, and 900 MPa in high strength steels, Figure 1 [1,5-9,25,26].

The mechanical properties of FRP bars are significantly different from steel bars. Its tensile strength is high, but its modulus of elasticity is small, and tensile diagram is linear, behaviour of concrete structure reinforced with FRP bars under loads, the stress-strain state and failure modes require to be investigated.

2 Methods

The finite element method was used for modelling of combined reinforcement by using ANSYS Workbench 2022 software. Non-linear properties of materials were also taken into account for calculating the structure. This makes it possible to obtain theoretical results that are very similar with experimental one and to compare them each other.

Design scheme of the beam is selected with a single-span, its full length is $l=1500$ mm, and effective length is $l_0=1400$ mm, height of the cross section is $h=200$ mm and width $b=150$ mm. Concentrated load is placed at a distance of $l_0/3$, taking into account that it is studied along the normal section (Fig. 2 and 3). Support of the samples are hinged, limited to X, Y and Z axes on one side, and Y and Z from the other support.

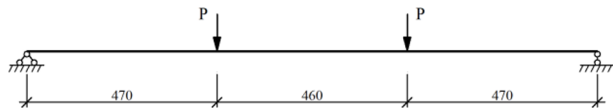


Fig. 2. Design scheme of the beam

Table 1. Program of numerical modeling

Beam ID	Series	Tension zone reinforcement	Compression zone reinforcement	Concrete grade	Research task
B1-3S12 (control beam)	Series 1	3Ø12 A-III	2Ø8 A-III	B20	load bearing capacity of the beam;

B2-1S12-2F10		1Ø12 A-III 2Ø10 BKA	2Ø8 A-III	B20	deflection corresponding to applied load; stresses in concrete and reinforcements
B3-1S12-2F12		1Ø12 A-III 2Ø12 BKA	2Ø8 A-III	B20	
B4-4S12 (control beam)	Series 2	4Ø12 A-III	2Ø8 A-III	B20	
B5-2S12-2F12		2Ø12 A-III 2Ø12 BKA	2Ø8 A-III	B20	
B6-2S10-2F12		2Ø10 A-III 2Ø12 BKA	2Ø8 A-III	B20	
B7-5S10 (control beam)	Series 3	5Ø10 A-III	2Ø8 A-III	B20	
B8-3S10-2F10		3Ø10 A-III 2Ø10 BKA	2Ø8 A-III	B20	
B9-2S10-3F10		2Ø10 A-III 3Ø10 BKA	2Ø8 A-III	B20	

For numerical modeling of the beam, a total of 3 series and 9 different reinforcement samples were selected, 6 of them (B2-1S12-2F10, B3-1S12-2F12, B5-2S12-2F12, B6-2S10-2F12, B8-3S10-2F10, B9-2S10-3F10) in the combined reinforced of steel and BFRP rebars, the remaining 3 samples (B1-3S12, B4-4S12, B7-5S10) are "control beam" with only steel rebars, which were accepted for comparison with beams used in combined reinforcement (Fig. 4).

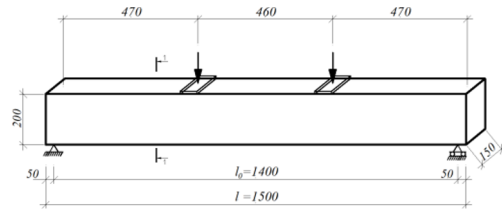
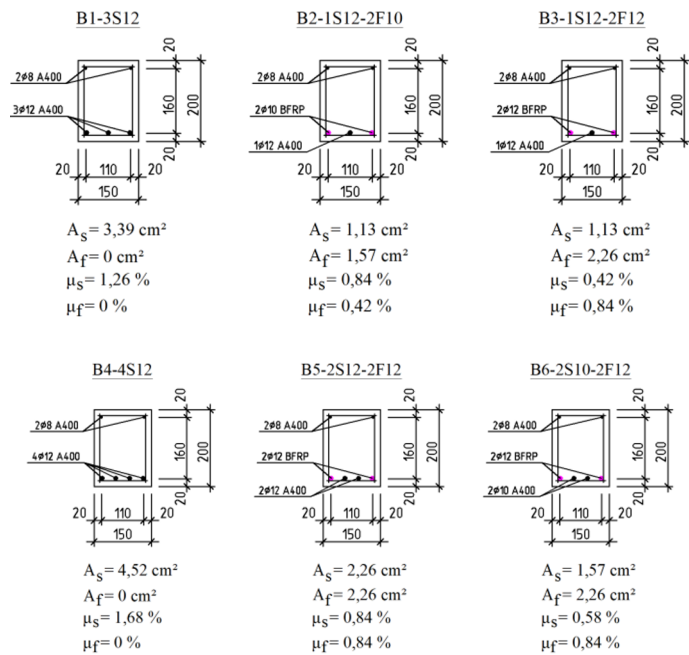


Fig. 3. Scheme of beam loading and support



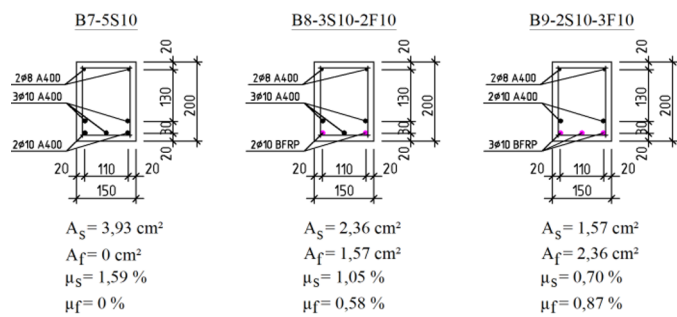


Fig. 4. Cross section of the beams

B1-3S12, B2-1S12-2F10 and B3-1S12-2F12 samples have single layer of 3 reinforcements in tension zone, in beams B4-4S12, B5-2S12-2F12 and B6-2S10-2F12 beams have single layer of 4 reinforcements and B7-5S10, B8-3S10-2F10 and B9-2S10-3F10 beams have 2 layers of 5 reinforcements which 2 reinforcements are installed deeper into beams and the remaining 3 bars located closer to surface.

Ø6 A240 grade rebar was used as transverse rebar in all beams, and distance of 50 mm was installed in $1/4$, and transverse rebars were not installed in the middle part $1/2$.

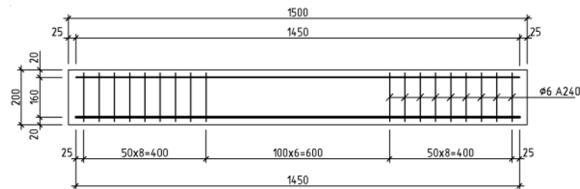


Fig. 5. Placing reinforcement cage in the beam

The sample was divided into 25 mm linear finite elements in 3D modeling using ANSYS. CPT215 function was used to study the nonlinear behavior of concrete using the Drucker-Prager method (Table 2).

Table 2. Nonlinear properties of concrete by the Drucker-Prager method

Parameter	Symbol	Value	Units
Modulus of elasticity	E	27000	MPa
Poisson's ratio	μ	0.2	
Uniaxial compressive strength	f_{uc}	20	MPa
Biaxial compressive strength	f_{bc}	23	MPa
Uniaxial tensile strength	f_{ut}	1.81	MPa
Tension cap hardening factor	R_t	1	-
Hardening parameter	D	30000	MPa ²
Compression cap location	σ_v^c	-15.33	MPa
Compression cap shape	R	2	-
Threshold for tension damage	γ_{t0}	0	-
Threshold for compression damage	γ_{c0}	0.00002	-
Tension damage parameter	β_t	3000	-
Compression damage parameter	β_c	2000	-
Nonlocal interaction range parameter	c	1600	mm ²
Over nonlocal parameter	m	2.5	-

Solid185 functions were used for modeling of reinforcement, and its mechanical properties are shown in Table 3.

Table 3. Calculated values of characteristics of steel and basalt reinforcement

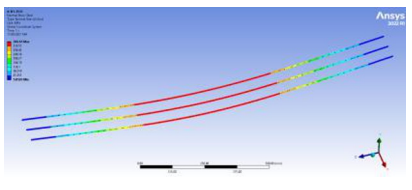
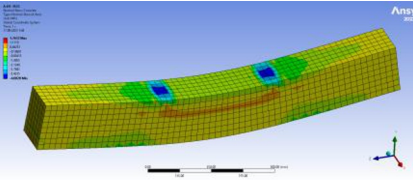
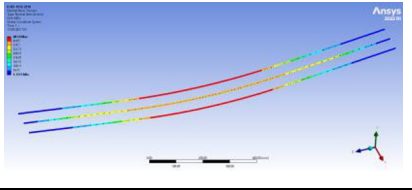
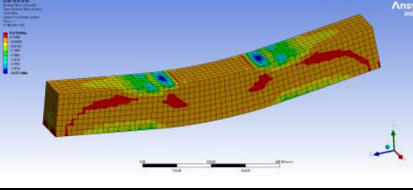
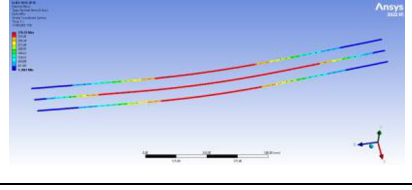
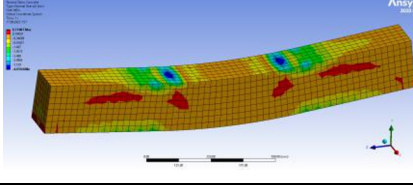
A400 steel rebar			BFRP rebar		
Parameter	Value	Units	Parameter	Value	Units
Elastic modulus	200000	MPa	Elastic modulus	50000	MPa
Poisson's ratio	0.3	-	Poisson's ratio	0.22	-
Yield strength	365	MPa	Yield strength	-	-
Ultimate strength	560	MPa	Ultimate strength	550	MPa

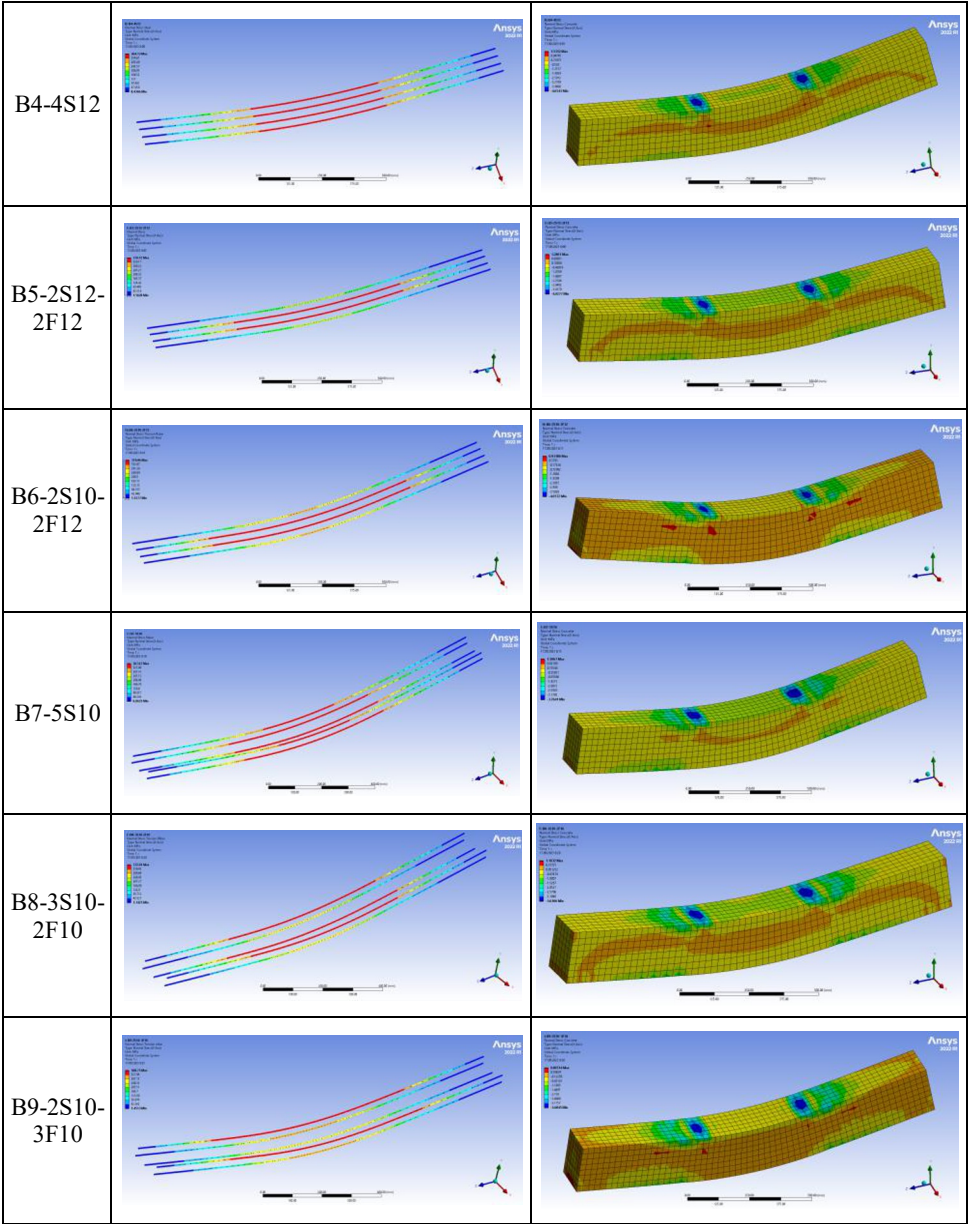
3 Results and discussion

As reinforcement ratios of the samples were balanced, failure modes in all samples were almost the same, it started with the failure of concrete in tension zone, then it was observed that steel bar stress in the tension zone reached its yield point. But there was a large reserve on BFRP rebar strength which means insufficient use of the strength of the basalt reinforcement. Crack opening width was larger than ultimate value.

When steel reinforcement stresses in the tension zone reached yield point $\sigma_s=R_s=365$ MPa, concrete and rebar stresses changed in different values depending on the amount of rebar used in the samples. Concrete and rebar normal stresses in all samples are shown in table 4. It can be seen from the table that at the point where concentrated load is placed on the beam, concrete stress in the compression zone is higher than other parts.

Table 4. Concrete and reinforcement stresses under destructive force

Beam ID	Rebar stress in tension zone, MPa	Stress in concrete, MPa
B1-3S12		
B2-1S12-2F10		
B3-1S12-2F12		



3Ø12 A400 steel rebar is installed single layer in tension zone of B1-3S12 beam, and when destructive force $P = 80.5\text{ kN}$, reinforcement stress reaches yield strength with the value of $\sigma_s = 366.6\text{ MPa}$, maximum stress of concrete in compression zone is equal to $\sigma_b = 4.08\text{ MPa}$, which means that its prism compressive strength is less than $R_b = 11.5\text{ MPa}$, maximum deflection is $f = 4.79\text{ mm}$.

There are 2Ø10 BFRP in the corners and 1Ø12 A400 steel rebar in the middle of tension zone of B2-1S12-2F10 sample. When destructive force is $P = 75.0\text{ kN}$, steel rebar stress is equal to $\sigma_s = 382.11\text{ MPa}$ and BFRP rebar stress $\sigma_f = 481.9\text{ MPa}$, concrete stress in compression zone $\sigma_b = 3.97\text{ MPa}$. But deflection has increased significantly $f = 21.33\text{ mm}$.

In tension zone of B3-1S12-2F12, there are 2Ø12 BFRP in the corners and 1Ø12 A400 steel rebar in the middle. When rebar stress $\sigma_s=376.76\text{ MPa}$, BFRP rebar stress $\sigma_f=357.19\text{ MPa}$ reduced compared to B2-1S12-2F10. Concrete stress in compression zone is $\sigma_b=4.06\text{ MPa}$ and deflection $f=17.14\text{ mm}$, destructive force is $P=76.5\text{ kN}$.

Destructive force reached the maximum value among all samples and was equal to $P=98.5\text{ kN}$ in B4-4S12. In this sample, only 4 steel rebars were used, and its stress was close to yield strength $\sigma_s=364.13\text{ MPa}$, and concrete stress was $\sigma_b=4.65\text{ MPa}$, and midspan deflection of the beam was equal to $f=5.24\text{ mm}$.

Tensile zone of B5-2S12-2F12 has 4 rebars in one layer, they are 2Ø12 BFRP and 2Ø12 A400. When destructive force $P=81.5\text{ kN}$, midspan deflection is equal to $f=9.77\text{ mm}$. Tension of steel rebar $\sigma_s=370.12\text{ MPa}$, and stress in composite reinforcement $\sigma_f=211.31\text{ MPa}$. BFRP rebar strength is significantly less than ultimate strength $R_f=550\text{ MPa}$. Stress in concrete is $\sigma_b=4.03\text{ MPa}$.

2Ø10 BFRP and 2Ø12 A400 were installed single layer in B6-2S10-2F12 beam, and when stress of steel reinforcement $\sigma_s=374.06\text{ MPa}$ reached yield point, BFRP rebar stress was $\sigma_f=291.7\text{ MPa}$. When destructive force was $P=77.5\text{ kN}$, maximum deflection in the middle of the beam was equal to $f=13.47\text{ mm}$, and concrete stress was equal to $\sigma_b=4.06\text{ MPa}$.

B7-5S10 control beam was considered, and 5Ø10 A400 were installed in tension zone, three of them in the outermost layer, remaining two rebars installed deeper into the beam. Destructive force is $P=80.5\text{ kN}$ and deflection is $f=6.42\text{ mm}$. Stress of reinforcement is $\sigma_s=367.67\text{ MPa}$, and concrete stress is equal to $\sigma_b=3.76\text{ MPa}$.

B8-3S10-2F10 has 2Ø10 BFRP and 3Ø10 A400 steel rebars installed double layer. When destructive force is $P=72.0\text{ kN}$, steel reinforcement stress is $\sigma_s=372.01\text{ MPa}$, BFRP rebar stress is $\sigma_f=256.69\text{ MPa}$, and concrete stress in compression zone is $\sigma_b=3.63\text{ MPa}$. Midspan deflection of the beam has increased significantly $f=11.49\text{ mm}$ than B7-5S10.

B9-2S10-3F10 beam there is a 3Ø10 BFRP in outermost layer and 2Ø10 A400 steel rebar in deeper. When the steel rebar stress is $\sigma_s=368.75\text{ MPa}$, basalt reinforcement stress $\sigma_f=291.93\text{ MPa}$ increased compared to B8-3S10-2F10. Concrete stress in compression zone equal to $\sigma_b=3.68\text{ MPa}$, maximum deflection $f=13.82\text{ mm}$, and destructive force $P=70.5\text{ kN}$.

Ultimate deflection $f_{ult}=l_0/130=1400/130=10.8\text{ mm}$. It can be concluded that the value of deflection in all beams with steel reinforcement is less than the limit value, but the deflection of beams with combined reinforcement exceeds ultimate deflection. Result show that there are problems in composite reinforcement, and it is acknowledged by the authors that solving this problem is one of the urgent and important tasks today. Also, in all combined reinforcement beams, when primary stress of reinforcement reaches the yield point, the strength in the composite reinforcement does not reach its ultimate value, and sufficient reserve remains in it, and this reserve of strength is not used due to the increase in deflection of the beams.

Table 5. Obtained results by virtual testing of samples

Beam ID	A_r/A_s	Destructive force, kN	Mid span deflection, mm	Steel rebar stress, MPa	BFRP rebar stress, MPa	Concrete stress in compression zone, MPa
B1-3S12	-	80.5	4.79	366.61	-	4.08
B2-1S12-2F10	1.39	75.0	21.33	382.11	481.9	3.97
B3-1S12-2F12	2.0	76 , 5	17,14	376.76	357.19	4.06
B4-4S12	-	98.5	5.24	364.13	-	4.65
B5-2S12-2F12	1	81.5	9.77	370.12	211.31	4.03
B6-2S10-2F12	1.44	77.5	13.47	374.06	291.7	4.06

B7-5S10	-	80.5	6.42	367.67	-	3.76
B8-3S10-2F10	0.67	72.0	11.49	372.01	256.69	3.63
B9-2S10-3F10	1.5	70.5	13.82	368.75	291.93	3.68

Due to the low elastic modulus of BFRP rebar, the value of deflections in combined reinforced concrete beams was higher than that of the control beam. As a result of increased FRP rebar area to steel reinforcement ($A_f > A_s$), stiffness of beam increased accordingly (Fig. 7,a). It can be seen from Fig. 7,b that even when steel and FRP bars are used in equal amounts, mid span deflection is large. In the tension zone, deflection of double layer rebar has reached greater than single layer rebar beams. If steel rebar places deeper into beam than FRP rebar, steel rebar is protects from corrosion, but its contribution to reduction of deflection is reduced (Fig. 7,c).

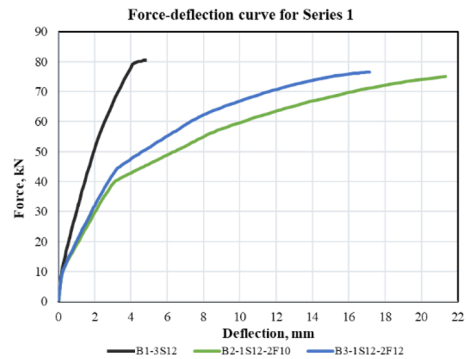


Fig. 7.a. Load-deflection relationship for beams (Series 1)

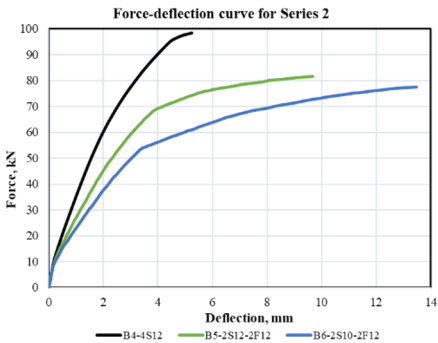


Fig. 7.b. Load-deflection relationship for beams (Series 2)

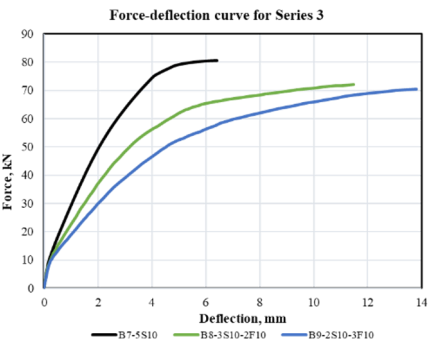


Fig. 7.c. Load-deflection relationship for beams (Series 3)

Concrete deformation and stress curve are nonlinear and do not obey any laws. In the Series 1, tensile concrete stresses of the control beam are greater than concrete stress of beam reinforced with combined steel and BFRP rebar. Moreover, concrete stress in specimens with combined steel and BFRP rebar began to develop earlier than in the beams reinforced with steel bar. In addition, concrete stresses of control beam greater than beams reinforced combined steel BFRP bar concrete stresses (Fig. 8.a, 8b, 8c).

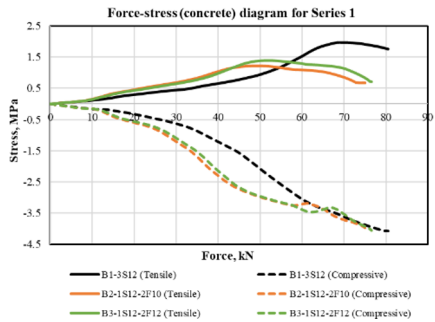


Fig. 8.a. Compressive and tensile stresses of concrete under applied load (Series 1)

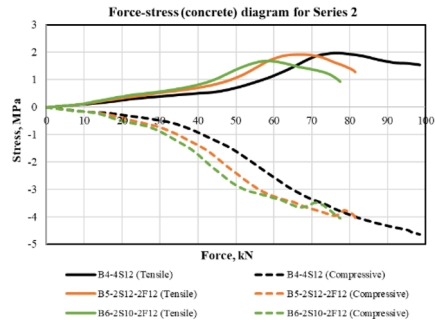


Fig. 8.b. Compressive and tensile stresses of concrete under applied load (Series 2)

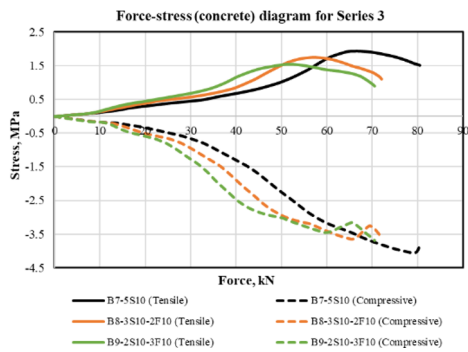


Fig. 8.c. Compressive and tensile stresses of concrete under applied load (Series 3)

When applied load is small, rebar stress gradually increases in beams reinforced only with steel bars (B1-3S12, B4-4S12, B7-5S10), force-deflection curve is linear until beam failure due to the increase of the load, and steel stress reaches yield strength, the sample loses its load bearing capacity. In the beams reinforced combined steel BFRP bar (B2-1S12-2F10, B3-1S12-2F12, B5-2S12-2F12, B6-2S10-2F12, B8-3S10-2F10 and B9-2S10-3F10), rebar stresses are small at the initial stage of loading and stresses are mainly taken only by concrete. As the value of the applied load increases, cracks are formed in the tension zone, as a result of which the load is received by the steel bars. Steel rebar stress increases rapidly in the stage before reaching the yield point, after which the deformation of steel reinforcements increases. While FRP bar stress gradually increase before steel reinforcements reach the yield point, basalt rebars begin to accept the loads after steel bars reaching yield strength (Figure 9a, 9b, 9c).

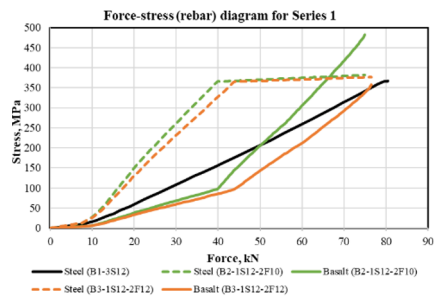


Fig. 9.a. Tensile rebar stresses under applied load (Series 1)

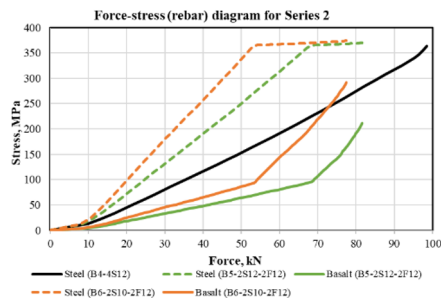


Fig. 9.b. Tensile rebar stresses under applied load (Series 2)

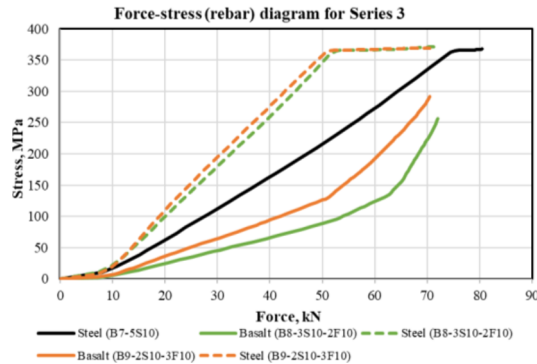


Fig. 9.c. Tensile rebar stresses under applied load (Series 3)

4 Conclusion

The following conclusions were obtained by the authors in order to modelling stress-strain state of combined steel-BFRP rebar reinforced concrete beams:

1. In all the samples, failure modes are almost the same, it starts failure of concrete in the tension zone, and at the same time, FRP rebar strength cannot be used sufficiently due to the fact that rebar stress in the tension zone reaches its yield strength. Moreover, it was observed that beam becomes unusable as a result of deformations exceeding the permissible values.
2. Concrete beams reinforced with steel bars are more stiff than combined steel-BFRP reinforced beams, and its deflection is less than ultimate values, but deflection of combined steel-BFRP reinforced concrete beams is higher than ultimate value.
3. Concrete stress in control beams was higher than beams reinforced with combined steel-BFRP bars. Also, concrete stresses in combined reinforced beams develops earlier and reached greater value than concrete beams reinforced with steel rebar.
4. As applied force to the beam less than 10 % of destructive force, load is mainly taken by concrete. This is why steel and BFRP rebar stresses were very small in the beginning stage of loading. As the load increased, stress-strain curve of steel bar became linear up to the yield point. But FRP bar showed its nonlinearity.
5. When a combined steel-BFRP reinforced concrete beam is subjected to load, initially stresses are taken by steel rebar then when it reaches yield strength, stresses are mainly taken by FRP bar.

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